Supplementary to "SpatialMuSC: A multi-scale perturbation-robust fusion model for spatial multi-omics domain identification"

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1. Data

1.1. Human lymph node dataset

This 10x Genomics-based spatial transcriptome–proteome dataset comprises 3,484 spots, with 18,085 genes measured in the RNA modality and 31 proteins measured in the ADT modality. Manual annotations provided by SpatialGlue [1] were used as reference labels. The anatomical domains are spatially discrete, making this dataset suitable for evaluating spatial domain identification, boundary delineation and cross-modality consistency in a complex human lymphoid microenvironment.

1.2. Mouse thymus dataset

This Stereo-CITE-seq spatial transcriptome–proteome dataset comprises 4,697 spots, with 23,622 genes measured in the RNA modality and 51 proteins measured in the ADT modality. The highly organized thymic architecture and distinct immune niches make this dataset suitable for evaluating whether integrative methods can preserve fine-grained spatial organization while capturing complementary transcriptomic and proteomic information.

1.3. Mouse spleen dataset

This SPOTS spatial transcriptome–proteome dataset comprises 2,568 spots, with 32,285 genes measured in the RNA modality and 21 proteins measured in the protein modality. The structured immune compartments of the spleen make this dataset suitable for evaluating spatial domain identification and multimodal representation learning in lymphoid tissue.

1.4. Adult human brain hippocampus dataset

This spatial-epigenome-transcriptome dataset comprises 2,500 spatial pixels, with 21,179 genes measured in the RNA modality and 56,614 chromatin-accessibility features measured in the ATAC modality. Manual annotations provided by MultiGATE [2] were used as reference labels. In contrast to the human lymph node dataset, the annotated domains are relatively continuous, making this dataset suitable for evaluating whether integrative methods can capture gradual spatial transitions and modality-consistent molecular variation in a heterogeneous neural microenvironment.

2. Evaluation metrics

To quantitatively evaluate the agreement between the predicted clustering and the reference annotation, we used eight complementary external clustering metrics, including adjusted Rand index (ARI) [3], normalized mutual information (NMI) [4], adjusted mutual information (AMI) [5], Fowlkes–Mallows index (FMI) [6], V-measure [7], F1-score [8], Jaccard [9] and completeness [7].

Let $Y = \{y_i\}_{i=1}^{N}$ and $\hat{Y} = \{\hat{y}_i\}_{i=1}^{N}$ denote the ground-truth and predicted labels for N spots, respectively, and let n_{ij} denote the number of spots simultaneously assigned to ground-truth class i and predicted cluster j. We first assessed pairwise clustering agreement using the adjusted Rand index, which corrects the Rand index for chance agreement:

$$\mathrm{ARI} = \frac{\sum_{i,j}\binom{n_{ij}}{2} - \left[\sum_i \binom{a_i}{2}\sum_j \binom{b_j}{2}\right]\Big/\binom{N}{2}}{\left[\sum_i \binom{a_i}{2}\sum_j \binom{b_j}{2}\right]\Big/2 - \left[\sum_i \binom{a_i}{2}\sum_j \binom{b_j}{2}\right]\Big/\binom{N}{2}} \tag{1}$$

where $a_i = \sum_j n_{ij}$ and $b_j = \sum_i n_{ij}$ denote the marginal counts of the contingency table. We next quantified the shared information between the two partitions using normalized mutual information:

$$\mathrm{NMI} = \frac{2I(Y;\hat{Y})}{H(Y) + H(\hat{Y})} \tag{2}$$

where $I(Y;\hat{Y})$ denotes the mutual information between Y and $\hat{Y}$, and $H(\cdot)$ denotes entropy. To further account for agreement expected by chance, we computed the adjusted mutual information:

$$\mathrm{AMI} = \frac{I(Y;\hat{Y}) - E[I(Y;\hat{Y})]}{\left[H(Y) + H(\hat{Y})\right]/2 - E[I(Y;\hat{Y})]} \tag{3}$$

where $E[I(Y;\hat{Y})]$ is the expected mutual information under random labeling with fixed cluster-size marginals. We also measured the geometric mean of pairwise precision and recall using the Fowlkes–Mallows index:

$$\mathrm{FMI} = \frac{TP}{\sqrt{(TP+FP)(TP+FN)}} \tag{4}$$

where TP, FP and FN denote the numbers of true-positive, false-positive and false-negative spot pairs, respectively. To summarize clustering quality from a conditional-entropy perspective, we used the V-measure:

$$\text{V-measure} = \frac{2hc}{h+c} \tag{5}$$

where h and c denote homogeneity and completeness, respectively. In parallel, we evaluated pairwise co-clustering consistency using the F1-score:

$$F1 = \frac{2TP}{2TP + FP + FN} \tag{6}$$

where the F1-score here is the pairwise F1-score computed from co-assigned spot pairs. We further quantified the overlap between correctly identified co-clustered pairs and all discordant pair assignments using the Jaccard index:

$$\mathrm{Jaccard} = \frac{TP}{TP + FP + FN} \tag{7}$$

Finally, we reported completeness to measure whether spots from the same reference class were assigned to a single predicted cluster:

$$\mathrm{Completeness} = 1 - \frac{H(\hat{Y} \mid Y)}{H(\hat{Y})} \tag{8}$$

where $H(\hat{Y} \mid Y)$ denotes the conditional entropy of the predicted partition given the reference annotation. For all eight metrics, larger values indicate better agreement between the inferred clustering and the ground-truth labels.

3. Supplementary Figures

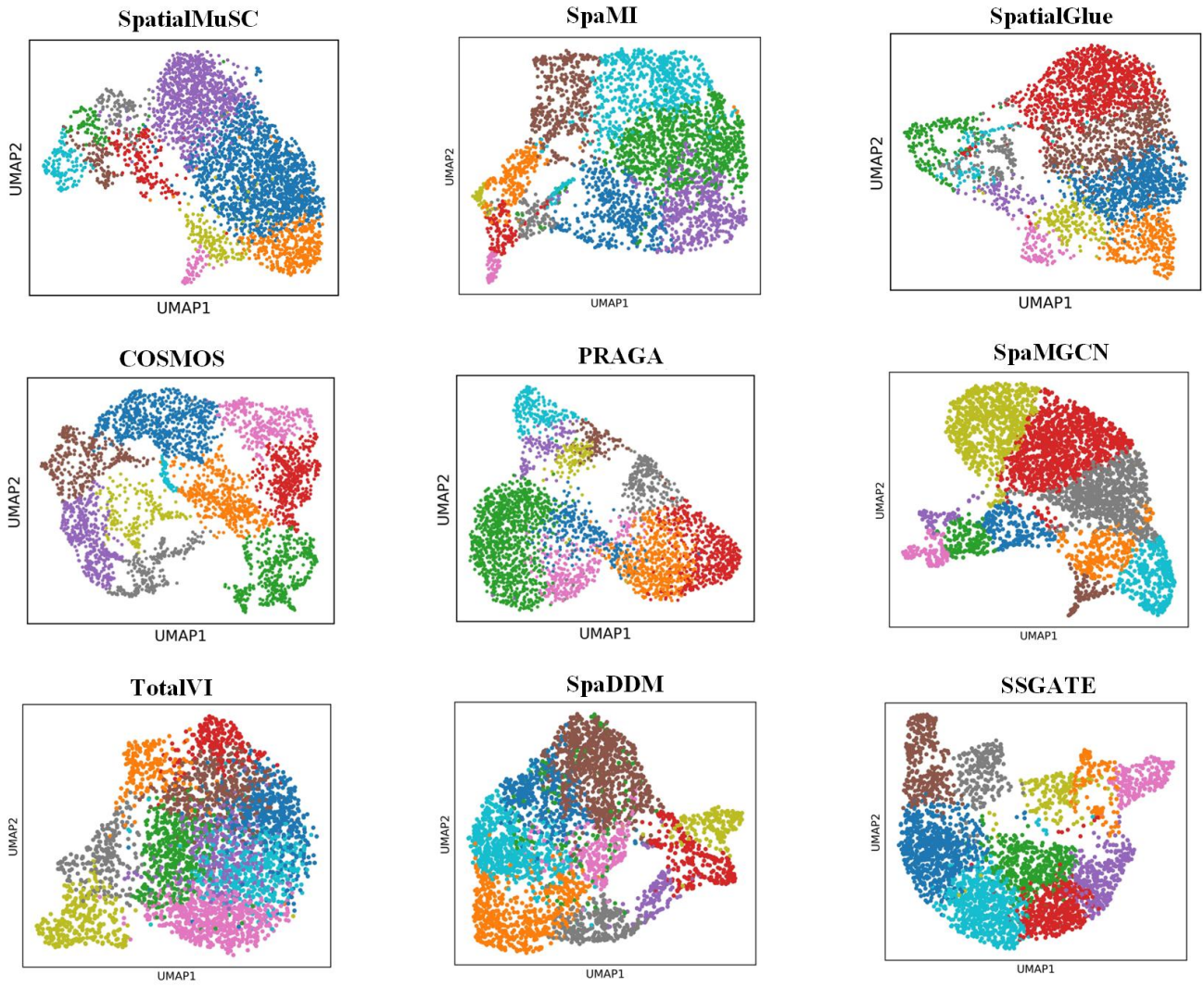

Figure S1: UMAP visualization of spatial domain predictions by SpatialMuSC and other methods on the human lymph node A1 dataset with a discrete spatial organization.

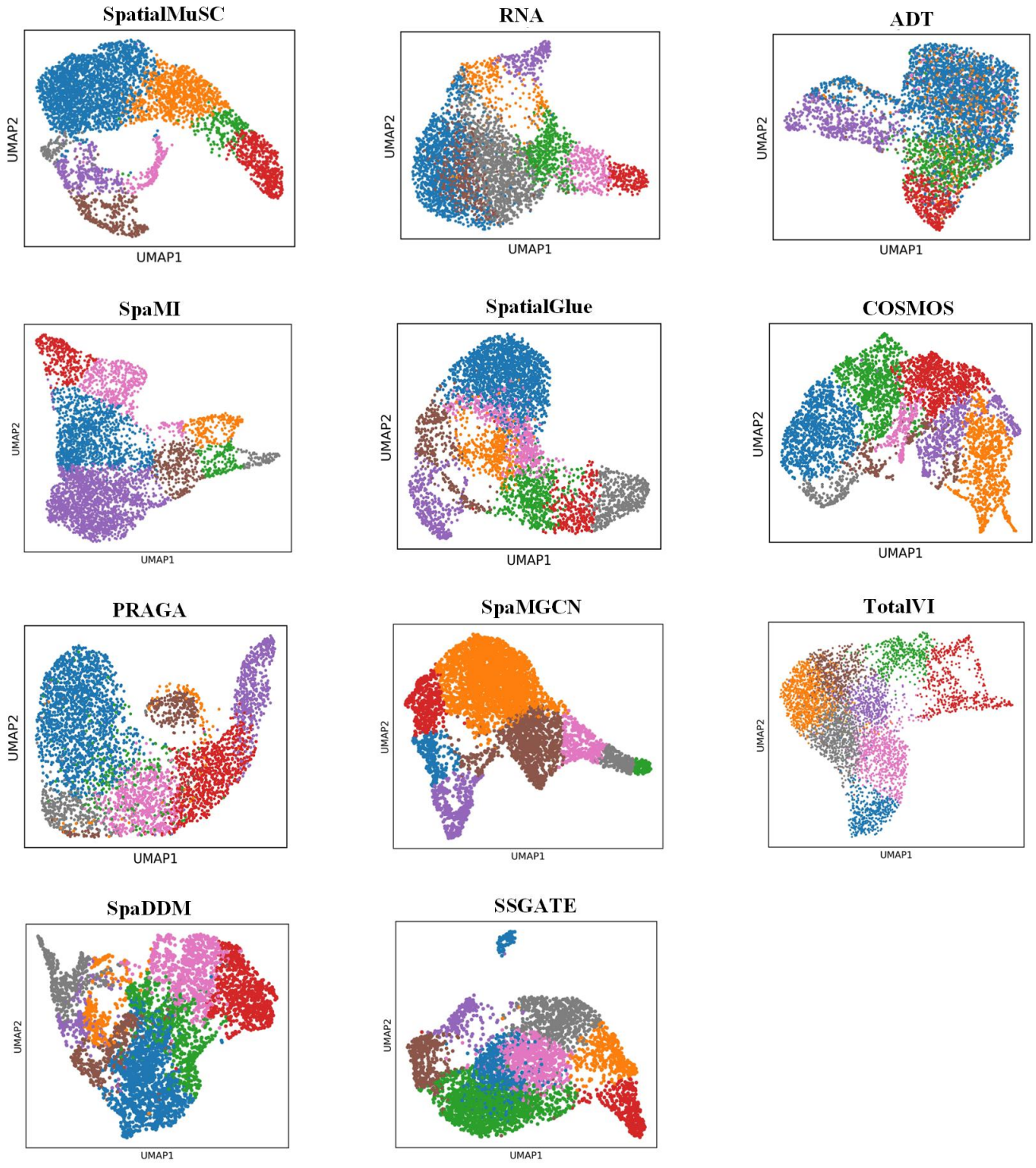

Figure S2: UMAP visualization of spatial domain predictions by SpatialMuSC and other methods on the mouse thymus dataset with a layered spatial organization.

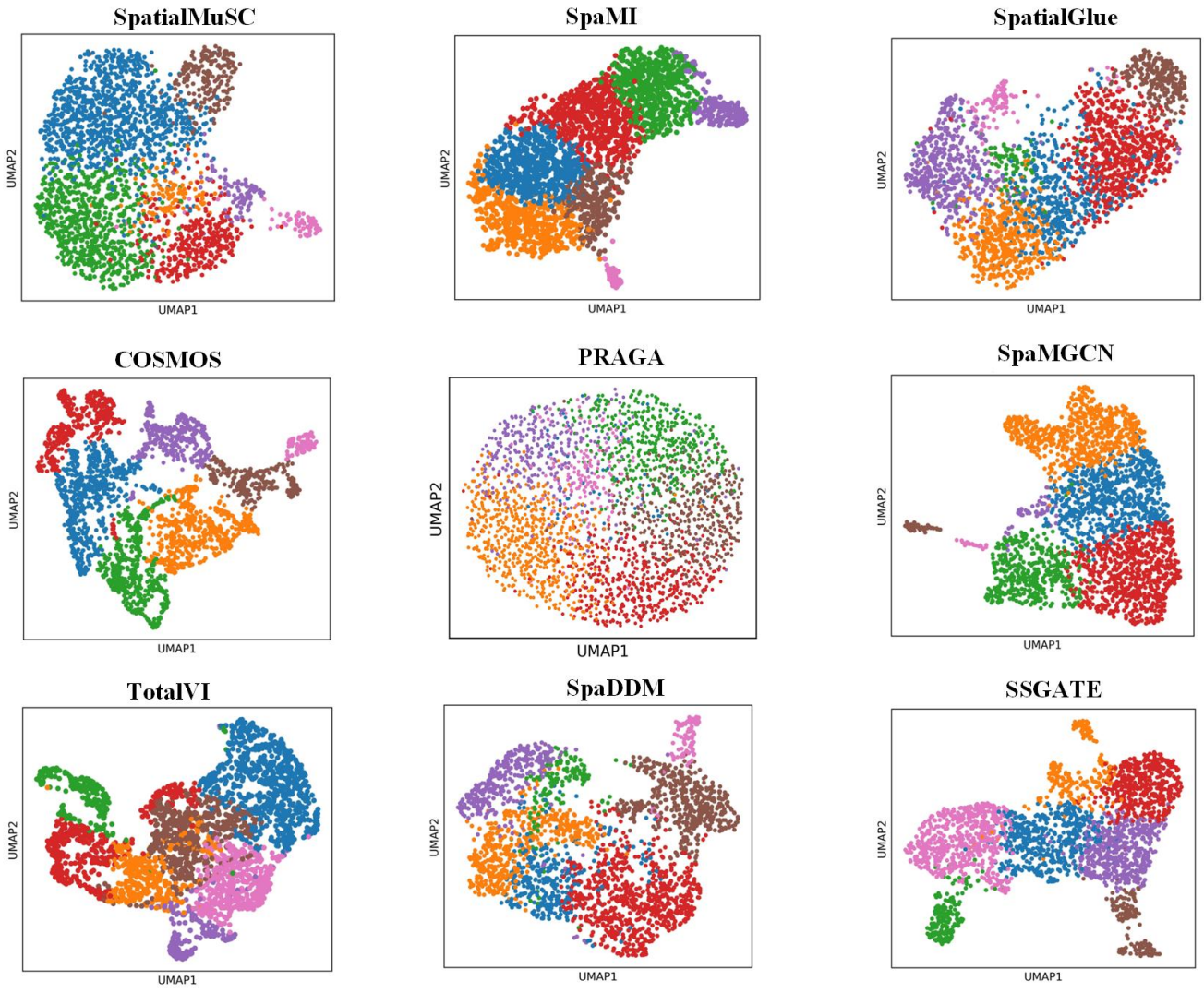

Figure S3: UMAP visualization of spatial domain predictions by SpatialMuSC and other methods on the adult human hippocampus dataset with a continuous spatial organization.

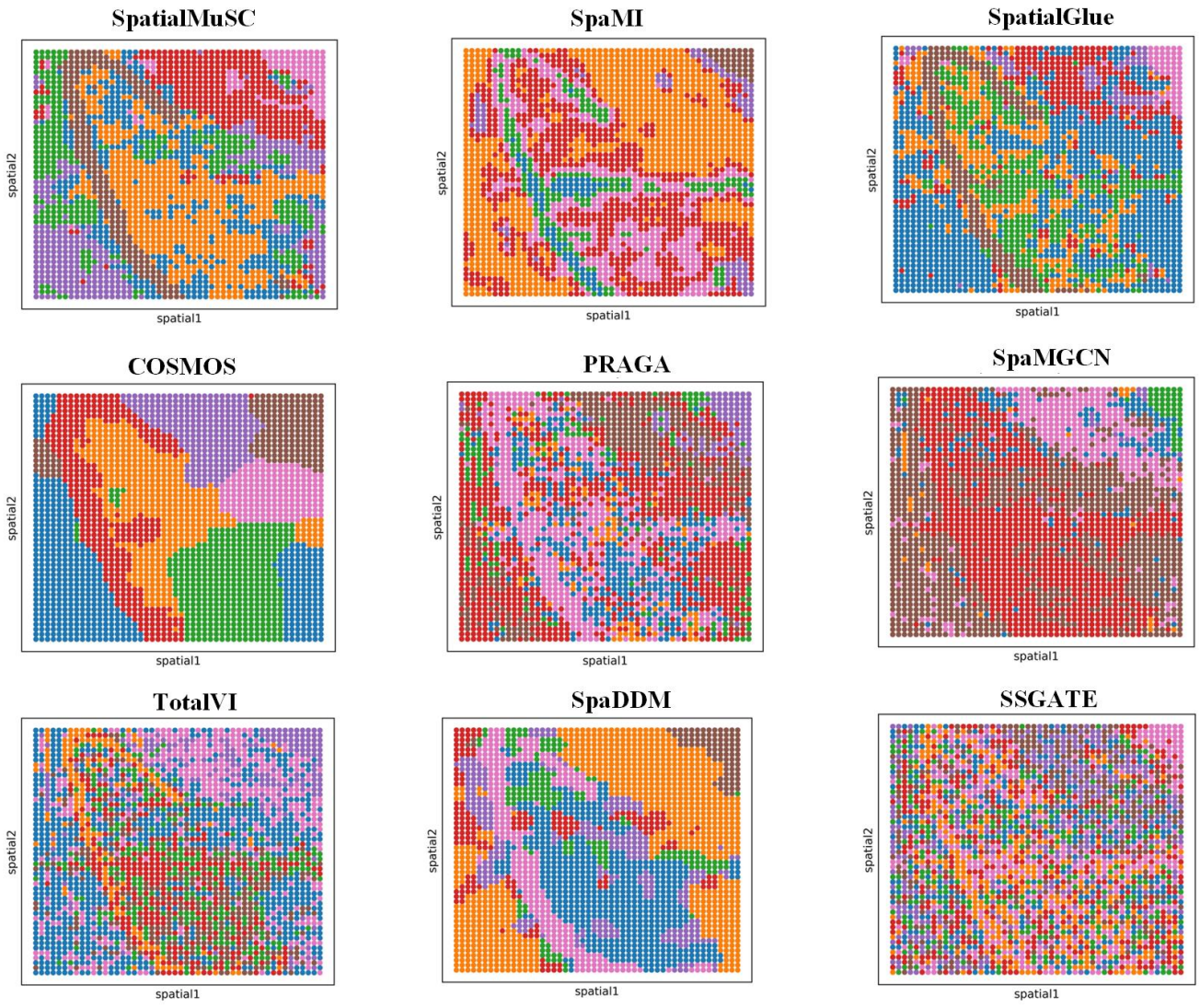

Figure S4: Visualization of spatial domain predictions by SpatialMuSC and other methods on the 50%-masked adult human hippocampus dataset with a continuous spatial organization.

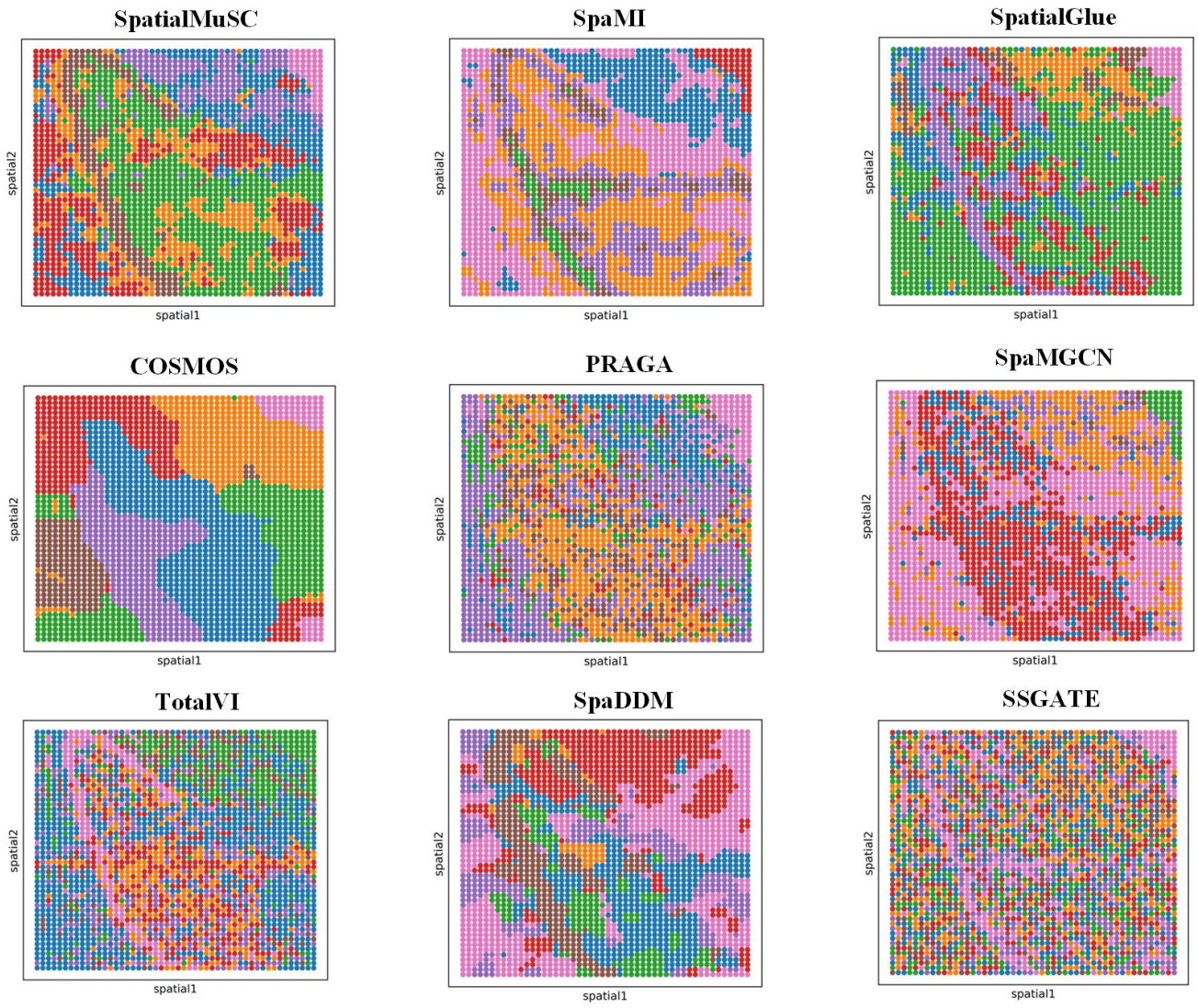

Figure S5: Visualization of spatial domain predictions by SpatialMuSC and other methods on the 70%-masked adult human hippocampus dataset with a continuous spatial organization.

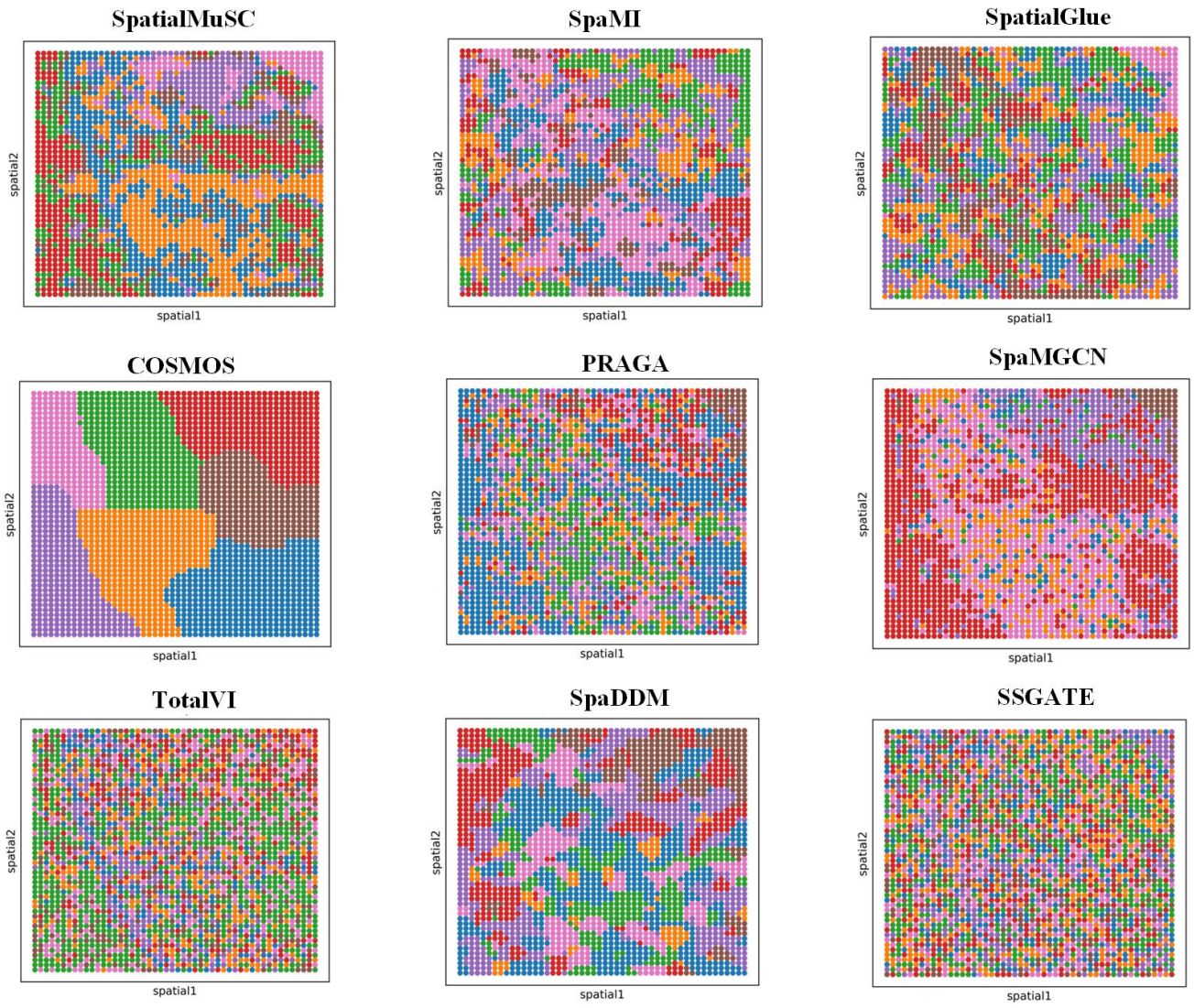

Figure S6: Visualization of spatial domain predictions by SpatialMuSC and other methods on the 90%-masked adult human hippocampus dataset with a continuous spatial organization.

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